

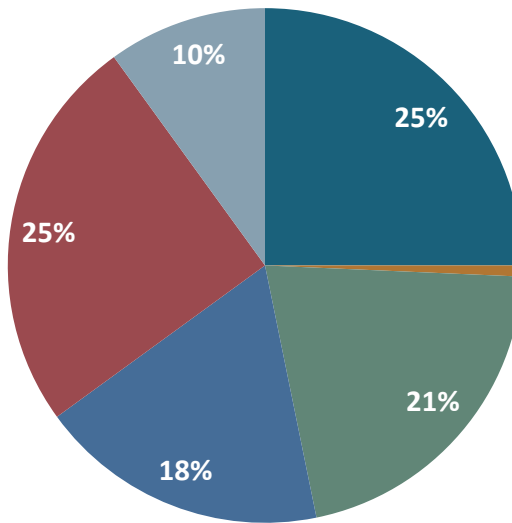
Portfolio Optimization

Minimum-Variance Allocation Across an 8-ETF Universe

Recommended Allocation

Recommended Allocation — Minimum-Variance Portfolio

■ AGG ■ EFA ■ GLD ■ QQQ ■ SPY ■ TIP ■ TLT ■ VNQ



Portfolio Impact

Minimum-variance optimization cuts expected volatility by roughly 27% versus naive equal-weighting — a deliberate risk-first trade-off that sacrifices some expected return for meaningfully smoother performance.

Metric	Recommended Portfolio	Equal-Weight Benchmark
Expected Annual Return	6.5%	8.5%
Annual Volatility (Risk)	7.1%	9.7%
Sharpe Ratio (rf = 4.0%)	0.36	0.47
Effective # of Assets	4.7	8.0

Why This Allocation Works

Diversification Logic

- Objective: minimize portfolio volatility across 8 asset classes spanning equities, bonds, gold, and real estate.
- Constraints: long-only positions; no single ETF exceeds 25%; weights sum to 100%.
- GLD's low correlation to equities (0.07–0.20) earns it a large 21% allocation despite its own volatility — the diversification benefit outweighs standalone risk.
- TLT is highly correlated with AGG and TIP (0.73–0.90), creating redundancy — hence its smaller 10% weight versus the 25% caps on the bond pair.

Correlation Snapshot

- GLD vs. SPY: 0.09 — strong diversifier, low overlap with equity risk.
- GLD vs. QQQ: 0.07 — near-zero correlation, ideal hedge against tech drawdowns.
- TLT vs. AGG: 0.90 — highly redundant; capped lower despite similar risk profile.
- TLT vs. TIP: 0.73 — overlapping duration exposure limits its allocation.

Correlations computed from 119 months of historical returns.

Important Disclosures & Next Steps

Disclosures

- This analysis is based on 119 months of historical ETF returns and does not guarantee future performance.
- Diversification does not eliminate the risk of loss.
- This material is for informational purposes only and does not constitute personalized investment advice.

Recommended Next Steps

- Review the recommended allocation and risk trade-offs with your advisor.
 - Confirm risk-free rate assumption and update if your situation has changed.
 - Schedule a follow-up to revisit weights as market data updates.
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